

The Role of MarineCadaastre.gov in Offshore Energy Planning

Christine M. Taylor
Department of Interior
Bureau of Ocean Energy Management
Herndon, VA
Christine.Taylor@boem.gov

Brian Smith and David Stein
Department of Commerce
NOAA Coastal Services Center
Charleston, SC
Brian.M.Smith@noaa.gov

Abstract— Many states along the U.S. East Coast are engaged in offshore wind energy planning through the Renewable Energy Task Force process led by the Bureau of Ocean Energy Management (BOEM). Participants from federal, state, and local government use the task forces as a venue to share information and minimize potential conflicts between existing ocean uses and offshore wind farm construction and operations. Task force participants often only make information available in static map form, making comparisons between data sets challenging. To address this, the North and South Carolina Renewable Energy Task Forces are collaborating with the MarineCadaastre.gov team to facilitate data sharing.

The National Oceanic and Atmospheric Administration (NOAA) and BOEM have been collaborating since 2007 on MarineCadaastre.gov, an integrated marine information system that provides jurisdictional, legal, physical, ecological, and ocean use data in a common geographic information system (GIS) framework. The MarineCadaastre.gov website was developed in response to a call for a marine information system to support alternative energy development on the outer continental shelf in section 388 of the Energy Policy Act of 2005. Although the website has been designed specifically to support renewable energy siting within the U.S. outer continental shelf, it is also being used for various other ocean-related efforts. The MarineCadaastre.gov website has three primary focus areas: Web map viewers and decision-support tools, a spatial data registry, and technical support and regional capacity building.

The North and South Carolina Wind Energy Task Forces are combining national-scale data from the website with information specific to their planning areas into simple viewers available in the Map Gallery section of the website. This approach allows users to assess and minimize potential conflicts of offshore wind energy development and operation with existing ocean uses. Multiple issues, such as fishing, shipping, and defense, are viewed and considered simultaneously, in one location, allowing members of the task force to see their own data in a wider context. This enhances the utility of the information that was previously compared using static maps. Users can turn data layers on and off and pan and zoom to specific locations in the planning area as questions arise. The North and South Carolina Wind Energy Task Force data viewers are a simple but powerful mechanism for identifying areas of potential incompatibility, enhancing communication between task force members, and increasing the overall effectiveness of the task force process.

The overarching goal of this work is to support the BOEM task forces in making sound decisions with the best available

spatial information to facilitate responsible siting of offshore renewable energy infrastructure. This presentation will provide an overview of the primary focus areas of the MarineCadaastre.gov project and its underlying technology, with particular attention given to the collaborations with the BOEM-led renewable energy task forces in North and South Carolina. Details on how these two BOEM task forces are incorporating use of the data viewers into their decision-making processes, the types of data being considered, and potential next steps for analyzing the spatial information in a group setting will be discussed.

Index Terms—Renewable Energy, Marine, Cadaastre, GIS, Marine Spatial Planning, Data-sharing, North Carolina, South Carolina, Task Force Meeting.

I. BACKGROUND

Human uses of ocean resources are increasing dramatically and outgrowing the laws, policies, and human resources needed to manage them. Development of offshore renewable energy is one example of these uses on a list that also includes commercial and recreational fishing, maritime commerce and trade, and national security operations, among others. These uses demand ocean space and increase the potential for use conflicts. At risk is the health of our ocean ecosystems and the benefits they provide to coastal communities and national economies [1].

MarineCadaastre.gov, also known as the Multipurpose Marine Cadaastre, is a single point of entry for discovering, downloading, viewing, and learning about a wide range of coastal and ocean spatial information that can be used for decision making and planning for any ocean use scenario. The MarineCadaastre.gov project, through its Web map viewers and spatial data portal is intended to address the need for accessible and authoritative data and science-based decision-support tools that support Coastal and Marine Spatial Planning (CMSP) drivers like offshore wind energy development. Its primary purpose is to serve as a planning and screening tool to inform decisions on ocean uses, and specifically to provide the spatial context needed to make decisions about where suitable areas exist for offshore activities and where to avoid development.

This ever evolving project attempts to meet the user needs and keep pace with available technology without breaking the budget allotted for it. It was envisioned in 2005 by the Federal Geographic Data Committee (FGDC) interagency group known as the Marine Boundary Working Group (MBWG). This working group's role is to create best practices for boundary creation and encourage collaboration among its members. It had long been suggested within the MBWG that a single web-based source for the digital map layers that represent the many legislated boundaries adjacent to and within the US marine environment was necessary. This need was primarily due to the general lack of digital spatial data available from the authoritative sources. Boundaries such as the Submerged Lands Act, Territorial Sea, Exclusive Economic Zone, marine sanctuaries, marine protected areas, or lease blocks were not readily available or viewable from a single viewer or download space. And many of these boundaries had been created incorrectly by other non-authoritative sources or not updated for years, and/or existed in multiple versions.

Most of the source agencies for Federal ocean and coastal geospatial data had representation within the MBWG. In 2005 the Energy Policy Act was amended to include oversight for development of offshore "alternative" (renewable) energy to the federal government's authority. Sec. 388 of the Act called for a "mapping initiative" for the purpose of planning for "alternative" energy projects, and called for the Secretary of Interior to initiate the project and work with the US Coast

Guard (USCG), NOAA, and the Navy [2]. The Secretary of Interior named MMS (now BOEM) the lead for the "mapping initiative." Multi-year funding made available through BOEM's Renewable Energy Office lead to an interagency agreement between BOEM and NOAA to create the MarineCadastr.gov project.

BOEM and NOAA are both major data providers for the project. BOEM is the project lead, and NOAA is the technical lead. The primary role of the data portion of the project is to provide "authoritative" ocean and coastal data. In this way, users can always be certain that they are downloading or displaying the most up to date version of the data and know it is from the source that is federally mandated to provide it. Through the MBWG, both agencies worked to help get the most up to date and most useful data from the other partner agencies such as USCG, USF&W, NGA, Navy, USGS, NMFS, EPA and DOE.

In the early years of the project the majority of the effort was focused on accessing the jurisdictional and navigational spatial data and the associated metadata from the federal authoritative sources represented within the MBWG and making them available as Environmental Systems Research Institute (ESRI) ArcIMS layers in a simple map viewer. Users had the basic ability to turn layers on and off, and print the resulting maps and legends. The original intent was to channel the authoritative data directly from the authoritative source's data servers, and not to have NOAA host the data from the other federal sources. Unfortunately, the technology to make

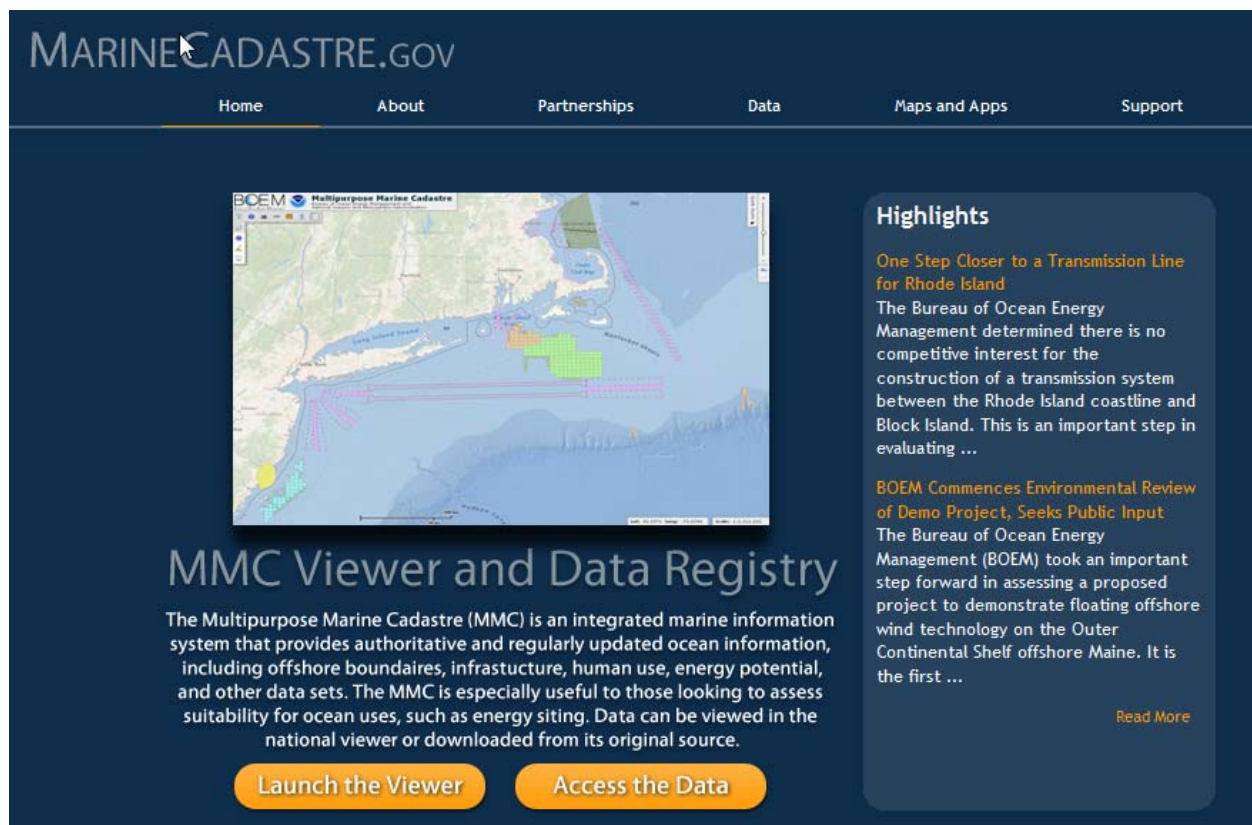


Fig. 1-MarineCadastr.gov Home Page [3]

that vision a reality was not yet readily adopted by the source agencies, primarily due to staff and budget constraints. The result was that more funding and work went into data creation, data harvesting and creating services for other agencies than was originally intended.

As technology advanced the project team developed the next generation of the viewer and special tools that used ESRI's ArcServer platform using the FLEX API technology. Additionally the team provided a single ESRI REST service for all the data sources that were being made available through NOAA. This allowed the data to be dynamically updated without updating the viewer each time. It also paved the way for allowing data from other REST services to be dynamically linked to the MarineCadastr.gov viewer. The MarineCadastr.gov page was also created to provide a non-agency-specific front end for the system. The project team has used a variety of venues such as conferences, working groups, special interest meetings, and offshore wind state task force meetings to increase awareness of the project for potential users, and solicit feedback on its utility and data holdings.

Based on user feedback the project team has included a large number of available data sets. A lot of effort has been spent investigating collaboration opportunities to include data from agencies that have not typically provided their data in a format easily used by GIS systems or that do not have the infrastructure to host the data themselves. For instance, the USCG made its AIS (Automatic Identification System) ship tracking data available by request only, and the requestor would be given a proprietary format file that had not been filtered at all, resulting in the requestor waiting for months to get a file that was too large to load and impossible to read. Given the increased demand for the AIS data, the MarineCadastr.gov project team invested some of the budget into creating filtered AIS data in a ESRI Arc10 geo-database format along with special analysis tools, free to anyone who wanted to use the data. This data was also used by BOEM to create the Atlantic coast AIS analysis, and has been used by a number of regions to create region specific analyses.

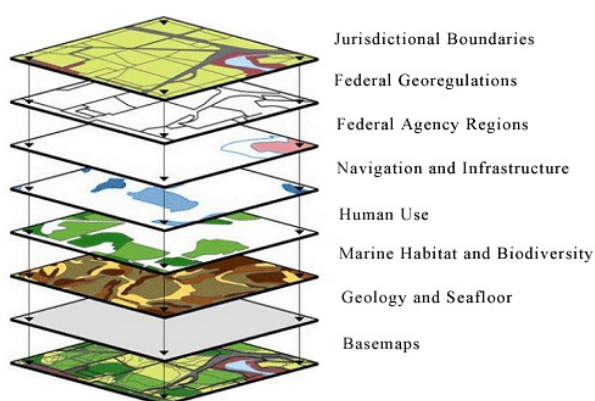


Fig.2: MarineCadastr.gov Data Themes

The primary tenets of the MarineCadastr.gov project have always been the following:

- Up to date authoritative-source ocean and coastal data
- Easy to use tools
- User support
- Stakeholder collaboration

II. SUPPORTING BOEM'S OFFSHORE WIND TASK FORCE MEETINGS

The BOEM Wind Energy State Task Forces have meetings as needed during the various stages of planning for offshore wind power in federal waters (3-200 nautical miles offshore). Each task force is set up for a particular state's interests even though the potential wind farms being discussed are usually wholly in federal waters but adjacent to a particular state or set of states. Other federal agencies also have a stake in the discussions depending on their particular mandates. The task force usually includes stake holders from the federal agencies, state and local government, and Native American tribal organizations. The meetings are held by BOEM in order to get input, sort out potential conflicts, and promote early coordination and planning with the purpose of avoiding last minute concerns or surprises. Maps are a large part of the discussion since areas of best wind generation locations are overlaid with areas of potential conflict with wind farm structures or construction processes. The primary objective of the task force in the earliest stages is to determine the areas that should be considered for development and further environmental analysis and eventual evaluation in an Environmental Assessment or Environmental Impact Statement under the National Environmental Policy Act (NEPA). By starting with the smallest areas of consideration, time and resources needed for NEPA analysis can be reduced.

Many of the task forces in the Mid-Atlantic region had already put together very comprehensive GIS layers meant for planning and task force meetings prior to the introduction of MarineCadastr.gov. In these areas the Environmental Analysis process had already started in the form of the Mid-Atlantic Wind Energy Areas outlined in the Secretary of Interior's "Smart from the Start" initiative. However, the maps were almost always presented as posters or MS PowerPoint files that could not be changed during the meeting or viewed after the meeting was adjourned.

With the newly adopted ESRI ArcServer platform, the MarineCadastr.gov team realized that they could take advantage of other technology such as ESRI's ArcGIS.com that could incorporate REST services and create on the fly special interactive maps, known as mashups, from a variety of sources, for specific meetings and without over-crowding the primary MarineCadastr.gov viewer. It also allowed the project team to make "non-authoritative" but generally accepted or "trusted" data available on an as needed basis within the individual mashups since the task force members could decide on which data layers they wanted to incorporate into the decision making process. This essentially allowed the marinecadastre.gov project to provide the lion's share of data

needed without breaking its own rules of only providing “authoritative data.”

ESRI’s ArcGIS.com, [5] which is available for free use, provided a web accessible basic map viewer that could incorporate REST services and individual GIS files to create a simple mashup map that could be made available to selected individuals, groups or everyone. The beauty of ArcGIS.com is that those data sets not yet incorporated into a more formal viewer such as the MarineCadastr.gov viewer, can be easily uploaded, identified, given a color or fill, and sorted. The end user can use this new map and turn layers on or off just like in the MarineCadastr.gov viewer. However, this can be set up and changed in a matter of minutes, thereby allowing the map creator to update maps as needed when new data is provided and change the maps for use before, during and after a meeting. As with the MarineCadastr.gov project, users can share new maps through email or web-based meetings or add temporary sketches, labels, descriptions, and comments.

In mid-2011, members of the MarineCadastr.gov project team, knowing of the need to create maps in North Carolina for the newly established NC Task Force, asked the BOEM Renewable Energy Office if they could assist in creating map mashups for the NC State Task Force meetings. A few conference calls and in person pre-meetings that included mapping staff and other data providers from the different stakeholder groups were held for information gathering purposes. NOAA started working on the mashup and as new data was made available they were quickly incorporated into the new ArcGIS.com map. The map was also linked to the

MarineCadastr.gov project through the Map Gallery page located on the website. Special criteria and analysis created for the individual meetings was provided and include data layers such as ship track analysis (AIS), fishing areas, DOD interests, hard bottom and living coral areas, state power grid, etc. Data collected by local universities or by state institutions were also made available. They included data such as shellfish beds, artificial reefs, fishing hotspots, electrical substations and special interest areas.

These map layers allowed the task force members to have access to data and maps that they would have only seen on the walls during the meetings. The ArcGIS.com maps provided a dynamic map interface that increased the task force member’s access to the data and increased transparency of the meeting information in general.

III. LEVERAGING SUCCESS

Additionally, discussions that occurred between the various data and map providers led to new collaborations to create data for inclusion into the main MarineCadastr.gov viewer, and helped to determine what the common data needs for the task force groups are regardless of which state is being represented. For instance, rather than having individual data sets for the ship tracking data created for each state, the USCG and BOEM collaborated to create an Atlantic coast version of the data. NOAA’s National Marine Fisheries, the Navy, and Duke collaborated to make data available that was originally used to determine the Navy’s impacts on marine mammal and turtle populations during their operations. As a result, multiple

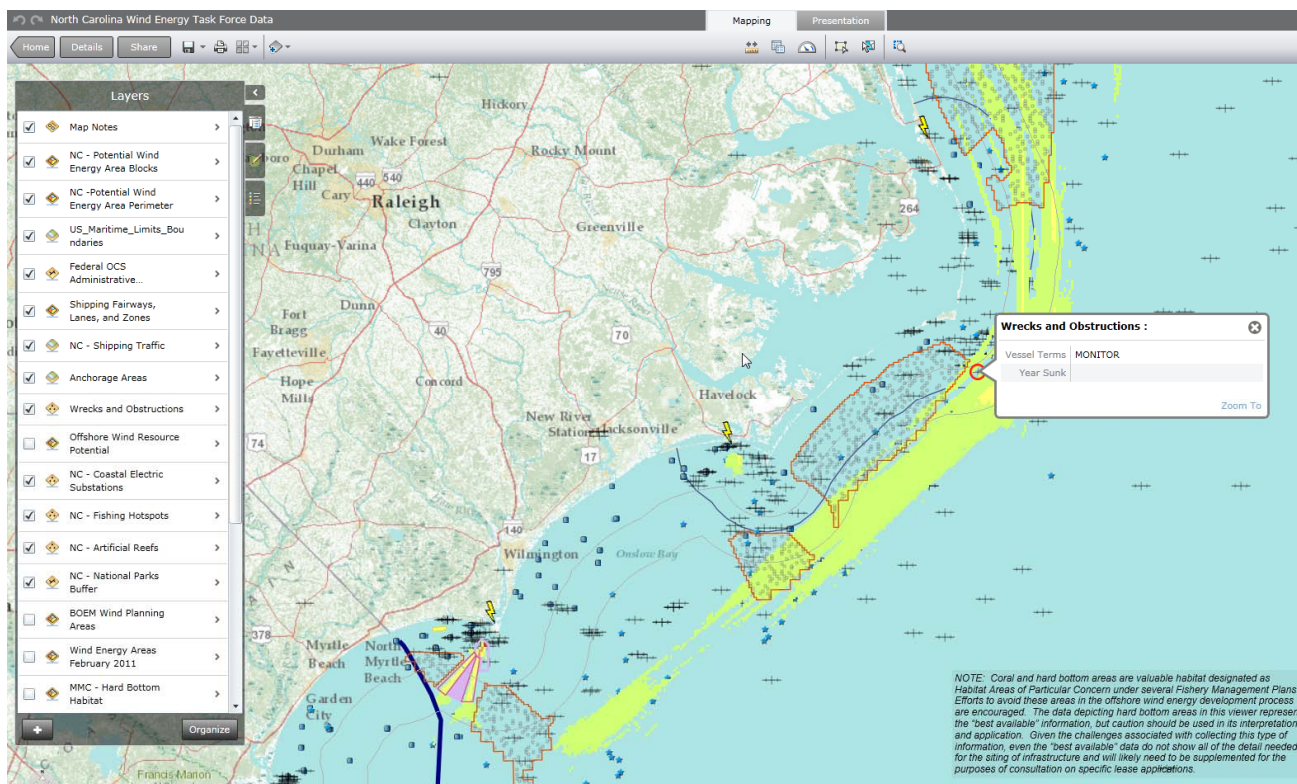


Fig. 3 –ArcGIS.com [5] Mashup for BOEM /North Carolina Offshore Wind Task Force Meeting

marine mammal and turtle layers are available on the MarineCadastr.gov site. Currently, BOEM is working with NOAA, USGS, and USF&W to produce a more complete analysis for the mid Atlantic of where certain bird species are likely to occur in the sparsely surveyed open ocean.

NOAA used the same approach that was used with North Carolina to put together a new map mashup for South Carolina. For the South Carolina Task Force, which had its first meeting in early March 2012, BOEM requested the task force members use the interactive map product shortly after the first meeting as a way to get everyone on the same page with what data were available and what the existing ocean uses looked like. This helped the task force members provide feedback on how things were being represented and let the members know if any data was overlooked, out of date, or misrepresented. South Carolina is the first task force to have this sort of map viewer available from the very start of the dialog. BOEM and NOAA staff involved with the MarineCadastr.gov effort are collaborating to find ways to increase the efficiency of the process using this spatial data during and in-between task force meetings as a mechanism to support the decision making process of identifying potential wind planning areas.

Currently, two NOAA offices and BOEM are making their data available as REST services, and a number of other federal sources are ramping up to make their data available in similar formats that can be pulled into multiple mapping systems. These services allow both the users and the providers to benefit. The data provider can make their data available as a single service and keep it up to date as needed, while the user can link to the data once, and know that the source will update it on the back-end, ending the constant checking and updating of data and reducing the uncertainty that always accompanied

that process. So, no matter what map viewer is being utilized, if they are all accessing the same REST service, the user and provider will know the data is the same.

Further, President Obama's July 2010 Executive Order 13547 establishing a National Ocean Policy (NOP) for the first time in the U.S. included a Framework for Effective Coastal and Marine Spatial Planning (CMSP) [4] and a call for the development of a National Information Management System to support its implementation. The core data and functionality of MarineCadastr.gov are now being leveraged into the development of a National Information Management System to support CMSP implementation in the U.S. It is also being leveraged within many regional portals and viewers that have been created with the intent of supporting the Regional Planning Bodies called for in the NOP.

IV. NEXT STEPS

The data found in the MarineCadastr.gov web mapping services are now being utilized in a number of different sources including two of the future regional planning bodies set up by the National Ocean Council. These include the Northeast and Mid Atlantic Regions which each have mapping data viewers that use MarineCadastr.gov services. The state of California is also working on a portal that access the same services, and ESRI is making its own maps from the same services using its own ArcGIS.com framework.

The far reaching influence that the MarineCadastr.gov project has had in the last few years has made it the first place that ocean planners may look for map layers showing areas of interest and conflicting areas that could influence the outcome of an offshore project. It is not meant to be the only and definitive source of information, but is meant to start the process of data gathering and initial map creation which was at

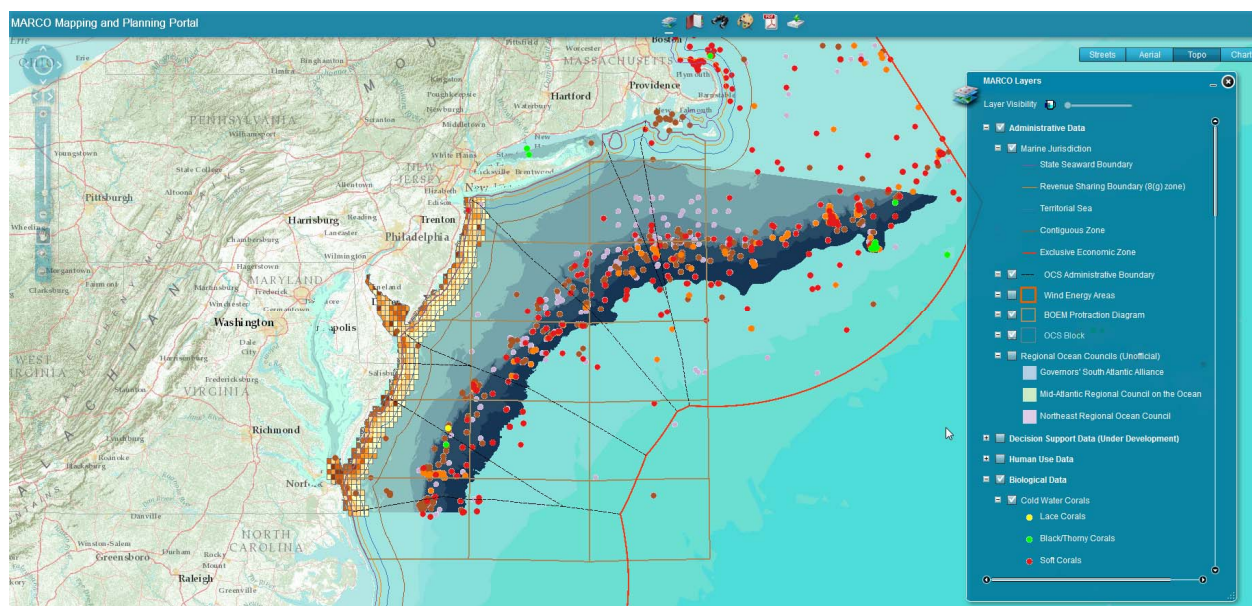


Fig. 4- Mixture of Regional and MarineCadastr.gov sourced data in the Mid Atlantic Regional Council on the Ocean (MARCO) map data portal.



one time done over and over again and took a long period of time to acquire data for each individual project. With the advent of the various regional mapping portals that access data from MarineCadastr.gov it has moved to the forefront of map data providers and has encouraged other authoritative data providers to attempt to make their data available in a geospatial format more easily accessed by the available map software and freeware used to make mashup map compilations. Suddenly the necessity for individual agencies and other

stakeholders to make their own policies, priorities, and concerns apparent in the face of offshore development planning, has spurred them to make their data available in a geospatial format available to the public. If these data sources do not have the resources to serve their own data, the MarineCadastr.gov team is often able to either create the

services or assist the other agency in making their data available using their own resources.

By making the data available as web services and linking to the authoritative sources whenever possible, the data is kept up to date by the data providers, and made available to multiple potential mashups, regional data viewers and portals. The mantra of “create once, use many times” is being realized through the MarineCadastr.gov project.

The MarineCadastr.gov team will continue to include national and large regional data layers available on the primary viewer and within the data registry for download. The team will also continue to support regional data portals and work with stakeholder groups such as the Offshore Wind Energy Task Forces and NOC Regional Planning Bodies to make the best and most up to date data available to the stakeholders, developers, and the public at large. The team will also make special tools available when possible to help with user research and stakeholder intercommunication efforts.

For more information, go to MarineCadastr.gov.

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